DATA SCIENCE & MACHINE LEARNING CAPSTONE

Reproducible Wildfire Exposure Screening for Residential Location Selection in Mainland Portugal

A reproducible historical screening tool, not a property-level wildfire forecast.

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Historical evidence compares broad areas—not properties

How can historical wildfire evidence help compare candidate residential areas in mainland Portugal?

The decision is national comparative shortlisting: apply one transparent method before investing time in detailed municipal, site, and property checks.

MAINLAND PORTUGAL

89,112
validated grid cells

1 km
sole analytical cell resolution

Scope boundary: historical comparative screening only—no property-level safety guarantee or purchase decision.

National coverage makes the method consistent

and supports honest temporal validation

NATIONAL FIRST

One method across the full range of mainland wildfire conditions

89,112 cells × 10 observation years supplied enough records for a strict temporal ML experiment without narrowing the buyer's national decision.

Consistent

The same definitions and processing rules apply across mainland Portugal.

Representative

The screening spans the country's broad range of historical wildfire conditions.

Testable

Temporal train and validation periods support an honest predictive experiment.

Reusable

The result is one inspectable national GeoPackage and QGIS presentation layer.

National screening does not replace municipal or property-level assessment.

One workflow keeps three evidence roles distinct

1
ICNF burned areas
Observed history 2016–2025

2
Canonical grid
1 km cells;
2 km context

3
Historical bands
0–1 / 2–3 /
4–10 years

4
ML experiment
2015–19 train
2020–21 validate

5
Official comparison
2020–2030
structural hazard

6
Present outputs
GeoPackage • QGIS
Figures • reports

Attempted regression context

CLC land cover • Copernicus DEM terrain • ERA5-Land JJAS climate • strictly prior fire history

The official ICNF structural-hazard layer is neither the regression target nor an accuracy benchmark.

Zero-heavy outcomes make low error deceptively easy

Exact zeros dominate the target

A regression that stays near zero can achieve a low average error while failing to rank many fire-affected cells near the top.

0

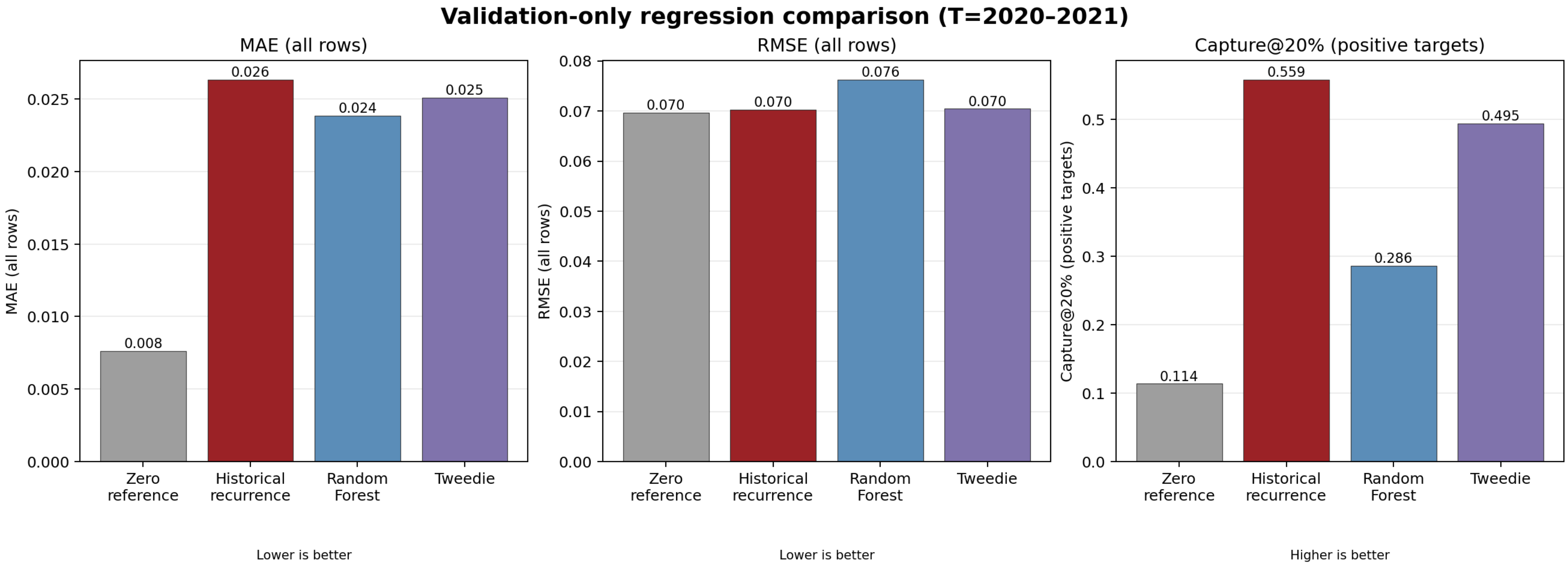
exact zero is a valid outcome

Three complementary questions

MAE	How large is the average absolute error?
RMSE	How strongly are larger errors penalised?
Capture@20%	How many positive-target cells appear in the top-ranked fifth?

Useful comparative prioritisation needs error and ranking evidence.

Validation rejected both predictive candidates



Neither candidate met the predeclared gate (lower MAE and RMSE, with capture@20% not lower). Historical recurrence is retained as descriptive screening evidence; no model advanced to final testing.

The ML models were not selected for prediction; observed historical recurrence was retained as the final, directly interpretable screening classification.

From buyer request to comparative location shortlist

1 Buyer request

Compare broad candidate areas before detailed property checks.

2 Reproducible lookup

Read observed historical exposure and the complementary official class consistently.

3 Appropriate next step

Prioritise local due diligence according to the evidence—not an automatic accept or reject.

Fictional example — not a property recommendation

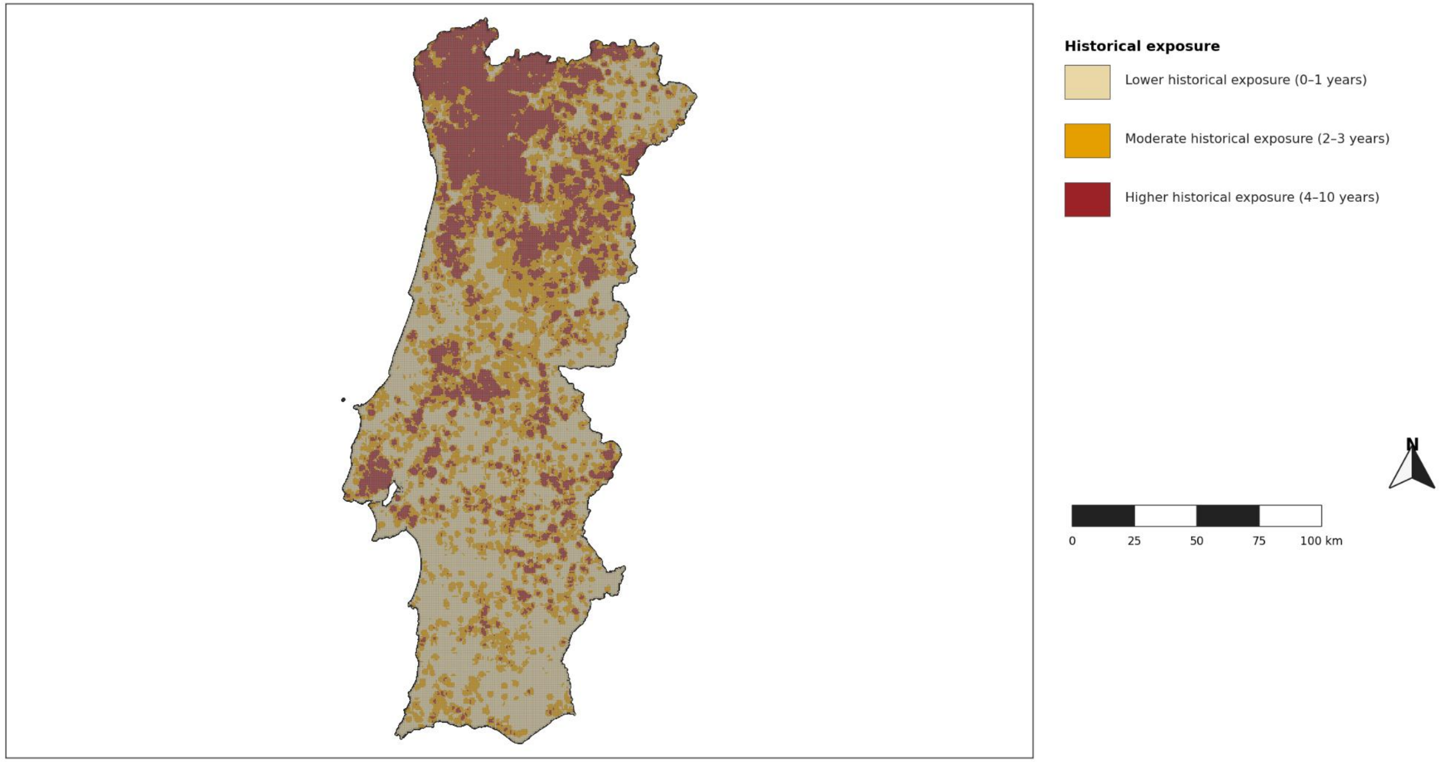
Candidate area	Historical exposure	Official structural hazard	Appropriate next step
Area A	Lower	Moderate	Prioritise detailed local checks
Area B	Moderate	High	Investigate prevention and access conditions
Area C	Higher	Very High	Require deeper specialist and local due diligence

The final classes summarise observed historical evidence consistently across the country. They support comparative shortlisting, not a property-level prediction or purchase decision.

Observed recurrence defines the final screening bands

Historical Wildfire Exposure Screening — Mainland Portugal

Historical exposure bands for 1 km mainland cells



EPSG:3763 | Sources: CAOP 2025 boundary; ICNF annual burned areas; ICNF structural hazard map 2020–2030.

1 km cells; recurrence measured in 2 km context; 2016–2025 evidence

Historical comparative exposure only; not a next-year forecast, property-level safety guarantee, or purchase recommendation.

OBSERVED 2016–2025 EVIDENCE

Lower

0–1 observed fire years

Moderate

2–3 observed fire years

Higher

4–10 observed fire years

1 km mainland grid cells with fire recurrence measured in a 2 km context.

Lower historical exposure does not mean safe.

Most cells fall in lower or moderate historical exposure

74.70% of mainland cells are in the lower or moderate historical-exposure bands

Lower
36,645 cells
41.12%

Moderate
29,919 cells
33.57%

Higher
22,548 cells
25.30%

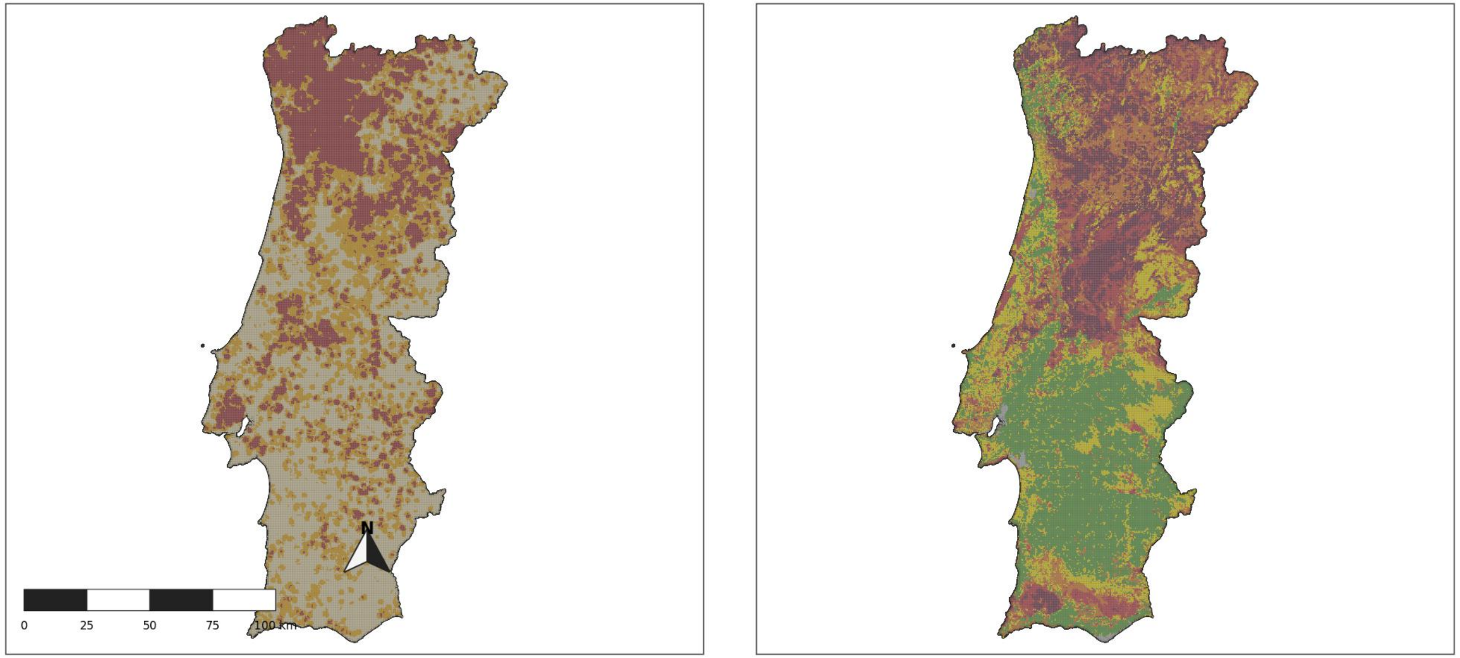
Complete national grid: 89,112 cells. Distribution is descriptive—not predictive performance or map accuracy.

The two maps answer different questions

Observed recurrence and official structural propensity are complementary

Historical Exposure and Official ICNF Structural Hazard — Comparison

Left: historical recurrence bands. Right: official ICNF structural-hazard class, summarized to 1 km cells.



Historical exposure

- Lower historical exposure (0–1 years)
- Moderate historical exposure (2–3 years)
- Higher historical exposure (4–10 years)

Official ICNF structural hazard

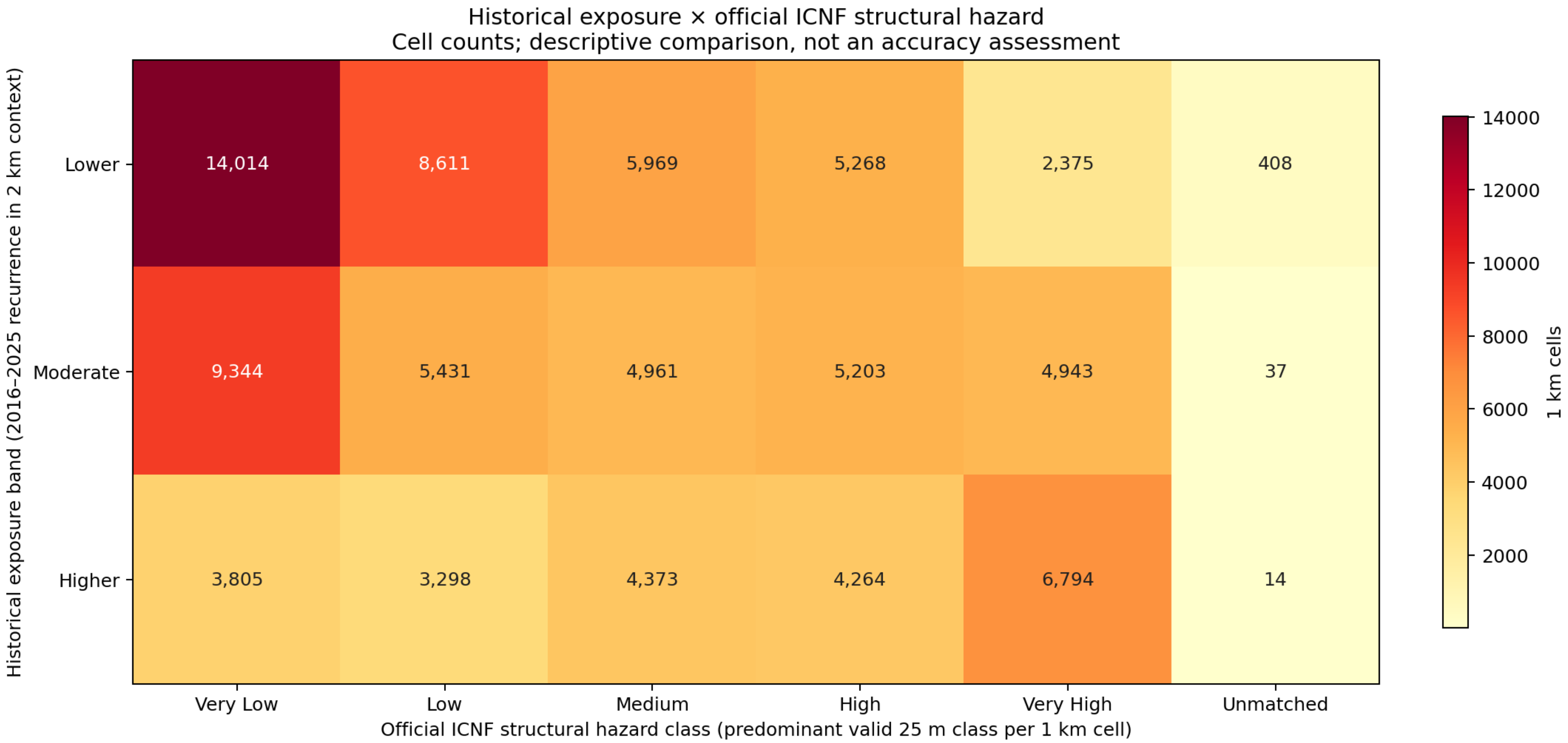
- Official very low structural hazard
- Official low structural hazard
- Official medium structural hazard
- Official high structural hazard
- Official very high structural hazard
- Official class unmatched

EPSG:3763 | Sources: CAOP 2025 boundary; ICNF annual burned areas; ICNF structural hazard map 2020–2030.

1 km cells; recurrence measured in 2 km context; 2016–2025 evidence

Historical comparative exposure only; not a next-year forecast, property-level safety guarantee, or purchase recommendation.

Correspondence is informative—not an accuracy score



43.59%

broad correspondence across matched cells

6,794	Higher × Very High
14,014	Lower × Very Low
3,805	Higher × Very Low
2,375	Lower × Very High

Descriptive correspondence only—not ICNF-map accuracy or model accuracy.

The project delivers screening with explicit limits

EVIDENCE-TO-USE CONTRACT

Validated ICNF evidence 2016–2025 → **1 km cells recurrence in 2 km context** → **Lower / moderate / higher historical exposure**

Use: broad location comparison and shortlisting; compare separately with official ICNF structural hazard.

Not a next-year forecast, property-level safety guarantee, wildfire-risk probability, or purchase recommendation.

CONCLUSION

1. Historical recurrence was the strongest validated evidence for comparative screening.

2. Random Forest and Tweedie were not selected because they did not improve sufficiently on validation.

3. The result is a reproducible, inspectable national GIS screening workflow.

ACTUAL DELIVERABLES

Python project + numbered notebooks • validated screening GeoPackage • portable QGIS project • figures, tables and validation reports • final PowerPoint presentation

Interactive layer: qgis/wildfire_exposure_screening_portugal.qgz

Likely assessor questions

Why all mainland Portugal?

The buyer's first comparison is national; local property checks follow after shortlisting.

Was the project successful?

Yes as a reproducible GIS screening workflow; no as an accepted next-year prediction project.

What was the hypothesis and result?

The seven-feature regressions did not beat the historical-recurrence reference sufficiently on validation.

Why is the classification still useful?

It directly summarizes observed 2016–2025 recurrence for consistent broad-location comparison.

Were the features sufficient?

They were appropriate for an initial reproducible experiment, but insufficient for an accepted next-year model.

Why retain recurrence if it is not prediction?

It was the strongest validated comparative evidence and remains transparently interpretable.

Central conclusion: no predictive model was selected; the final deliverable is historical comparative screening.